BASANT BHATTARAI

SENIOR DATA & AI ENGINEER / PLATFORMS, DATABASES & AGENTIC SYSTEMS

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PROFILE

Senior data and AI engineer with 9 years in software engineering, six of them owning production data platforms. Open-source first — Kafka, Spark, Iceberg, Trino, dbt and Airflow on Kubernetes — across ingestion, transformation and analytical serving, plus the databases underneath: modeling, replication, backup and recovery, upgrades. Designs and runs the agent workflows on that same stack — LangGraph, Google ADK and MCP tool servers — so model output lands as typed, reviewable rows rather than a transcript. Track record: 40% lower infrastructure cost, 50% faster processing, 60% faster analytics delivery.

EXPERIENCE

Senior Data Engineer | UXCam

Feb 2024 - present

- Own reliability and governance for a platform handling 15 TB+ and 100M+ events daily: freshness and latency objectives per dataset, 99.9% uptime, incident-ready runbooks, reviewable data contracts, a named owner for every published table, and the SOC 2 criteria met for the platform and its infrastructure through every audit.
- Design and operate the production agent workflows — LangGraph and Google ADK orchestration, MCP tool servers, retrieval and evaluation — against the existing platform, so model output lands as typed, provenanced rows rather than a transcript: analytics delivery 60% faster with 75% less analyst effort, review and fallback paths kept.
- Lead architecture and operational review across ingestion, transformation, storage, analytical serving, observability and recovery, without treating any single tool as the architecture.
- Run capacity and cost work as a design input rather than a monthly report, using storage layout, partitioning and compaction cadence as the primary levers.
- Mentor 5 engineers through pairing, code review and written design feedback; team delivery time down 30%, with the emphasis on making system ownership transferable.

Data Engineer | UXCam

Feb 2020 - Feb 2024

- Built and operated batch and streaming pipelines, then took on the data models, orchestration, storage and analytical serving paths around them.
- Rebuilt the core ETL layer in Spark and PySpark with Airflow orchestration, cutting processing time for 5 TB+/day by 50% and unblocking real-time product dashboards.
- Made change and recovery safe by treating schema evolution, replay, late-arriving data and backfills as designed interfaces with idempotent loads and explicit watermarks, not as emergency procedures.
- Tuned Trino, ClickHouse, Citus and TimescaleDB serving 100M+ queries/day for a 50% p95 latency reduction through partitioning, clustering and model redesign.
- Led the Kubernetes migration of the data infrastructure: 40% lower infrastructure cost, 99.9% uptime, and autoscaling that held under peak load.
- Worked across European and US time zones to turn product questions into maintainable datasets.

Project Leader | SVCET, India

2019 - 2020

- Led 4 developers building an intelligent dialogue system on an NLP and deep-learning stack, reaching the client's acceptance threshold for intent recognition and delivering 2 weeks ahead of schedule.
- Coordinated requirements, planning and review across academic and industry stakeholders, and introduced the code-review and version-control discipline the department kept afterwards.

Software Developer | SV Technology, India

2017 - 2019

- Built Python and FastAPI backend services carrying 10K+ users at 85% test coverage.
- Designed and maintained PostgreSQL schemas, stored procedures and reporting paths, and set up CI with Jenkins alongside the early pipeline work that moved me into data engineering.

SELECTED OPEN SOURCE

- **spark-sql-kinesis** (maintainer): Kinesis source and sink for Spark 4.2 on AWS SDK v2, on Maven Central; an idle shard costs one GetRecords call, not a multi-second floor per batch.
- **evalproof**: LLM evaluation with judge calibration and CI gates that fail only on drops that are both large and significant.
- **governed-agent**: a LangGraph and MCP agent whose answers become state only after schema, policy, grounding and review.
- **distributed-db-lab**: Citus sharding, online rebalance and tenant isolation, measured against TimescaleDB and replicated ClickHouse. Also streaming, real-time ML, retrieval, data contracts and GitOps delivery. github.com/technobasant

CAPABILITY MAP

Platform ownership

Consumer-facing reliability, governance, recovery and operational readiness.

Python, SQL and Scala; Kafka and Kinesis ingestion (maintainer of an open-source Spark 4 Kinesis connector); Spark/PySpark processing, including the Spark 4 migration; Airflow orchestration; Iceberg, Hive Metastore, dbt and lakehouse modeling; Kubernetes, Docker and Terraform delivery; contracts, lineage, cost and recovery.

Database lifecycle

Modeling, replication, failover, backup and PITR, upgrades and query performance.

PostgreSQL, ClickHouse, Trino, Citus, TimescaleDB, Redis and MongoDB; transactional and analytical modeling, partitioning, replication, backup, restore, high availability and query performance.

Agentic systems

Model output that lands as typed, reviewable rows rather than a transcript.

LangGraph, LangChain, Google ADK, MCP servers, Pydantic, Milvus and MLflow; bounded tools, retrieval, evaluation before rollout, typed output, provenance, human review and honest fallback paths; applied-ML evaluation of OCR and short-text classification against a fixed scorer.

Technical leadership

Design review, mentoring, runbooks, and making system ownership transferable.

Architecture and operational review, mentoring, incident-ready runbooks, service and freshness objectives, observability with Prometheus and Grafana, and distributed delivery across time zones.

Cloud

AWS: S3, EKS (IRSA), Lambda, Glue, Athena, Redshift, Kinesis, EMR | **GCP:** BigQuery, Dataflow, Dataproc, Datastream, Pub/Sub, Cloud Composer, GKE, Vertex AI | **Azure:** Synapse, Data Factory, Databricks, ADLS, Event Hubs, AKS, Azure ML

EDUCATION AND LANGUAGES

Bachelor of Technology in Computer Science and Engineering

JNTUA College of Engineering, Anantapur, India | 2015 - 2019

English: C1 | Nepali: native

Selected work, writing and case studies: basantbhattarai.com.np/work/